

Aspects of circuit protection

If you have watched the movie *Interstellar*, or are aware of the infamous Murphy's Law, then you know that, anything (bad) that can happen, will happen. Circuit faults can be generated internally because of problems in the system, or can come from external sources. The basic aim of circuit protection is to safeguard against loss of functionality, or degradation in performance, without affecting the core functionality of the system. It has three key elements: protection against mains voltage and its variations, protection against lightning and high voltage transients, and protection against ESD.

Internal problems can be due to over-current events such as a short circuit, or an overload current, and over-voltage events such as short-duration transient voltage spikes or longer duration over voltages. These faults can negatively impact sensitive electronic circuits, if not dealt with on time. External problems can include lightning interference, that induces surges inside the electronics; these travel from outside conductors on the grid to the inside of buildings and homes.

Most products need all three protection schemes, and the intensity of protection varies according to application. For example, if you are developing telecommunication products, your focus will be on mains and lightning. However, if it is a consumer product, the focus will be on ESD.

Srinivasa Moorthy, director, D4X Technologies Pvt Ltd, says, "Many times designers worry about the protection aspect after the design is complete and find out the problems only when the product fails in compliance tests. The key is to understand the compliance and protection needs. It is usually a matter of deriving protection parameters from the standards and implementing protection as part of the design, not as an afterthought. In my opinion, the most forgotten protection is the protection of communication connectors of the product against ESD."

Abhinav Patil, field applications manager - key accounts, Analog De-

Reddit in the house

u/cynar on r/AskElectronics says,

There are four main things to watch out for:

1. *Reversed polarity on the power supply.* Swapping the positive terminal for the negative will kill many circuits. A diode will work but is inefficient due to both voltage drop and current loss. A p-type metal-oxide semiconductor field-effect transistor (MOSFET) is the best option, but is expensive and needs to be matched to your circuit to an extent.

2. *Shorts/excess current.* A short to ground can cause a fire in some cases. But even an overloaded circuit can either heat components and connectors or destroy these outright. Best solution here is a fuse or polyfuse (a fuse that resets itself once the problem is cleared). This might not save the chip but it should stop a fire. Again, the value needs to be matched to your circuit.

3. *Over voltage.* Not generally a problem, but can be in some setups, especially when dealing with spikes. A cap on the input power will smoothen out any short spikes, while a flyback diode will stop spikes from motors coming back into the control circuit. Sustained over voltage is best dealt with a zener diode between the positive and negative terminal. This should be valued so that it only breaks down when the voltage is too high.

4. *Static.* This is the most painful process in this regard. Your body can build up a huge static voltage in some situations, of the order of 10,000V or more. This will fry many components, even if only applied for fraction of a second. The best protection, in such a case, is pre-emptive grounding of yourself. Touch a grounded metal object before working on your circuit and give the case a good ground if you can. Also, keep components in electrostatic bags until use, if you can (particularly transistors or transistor based stuff).

u/MagnusDre on r/AskElectronics says,

There are a few more things to be looked out for:

On the power bus, there should be reverse polarity protection both on the power input side at DC plug/headers and circuit side rails to ground and over the regulator. Most of the magic smoke I have seen came from reverse polarity, so I wish more people would use reverse polarity-protection diodes and mark the polarity really clearly.

Ex-rated systems use controlled impedances and zener diodes to keep maximum energy in a sensor under control.

While working with digital signals, use isolation devices like optoisolators to eliminate ground loops and massive all-system failure, especially in systems where these cross boards have long wires or go into sensitive converters.

For all devices, as a rule, high voltages should be out of reach of fingers always. This also applies to high energies, like the energy stored in a supercap, just to be on the safe side.

vices India, feels that design engineers need to perfect their workflow and not skip any aspects to ensure an optimum protection scheme. He explains, "There are certain international standards that define the tests an electronics system should pass to be certified as an EMI/EMC robust system, something essential to most products made for the consumer market. These tests include ESD, electrical fast transients (EFTs), surge, radio frequency interference (RFI) and others. The design engineer needs to have knowledge of these standards along with the operating conditions of the product, interference sources and mechanisms through which the interference can get coupled into the system, and the impact of the interference on the system."

Patil adds, "Overlooking any one of these aspects can result in failure to design a truly robust system. Designers also sometimes fail to anticipate non-ideal operating conditions and human errors (like miswirings). So a system that works well in the lab under ideal conditions may fail when installed in the field. Yet another mistake designers may commit is to design the protection scheme only for transient events, while ignoring possible steady state conditions (like high AC/DC voltage on the signal or ground path)."

What is new in this field

Following the trend of going small in electronics, circuit protection has also taken strides in that aspect. New fuse technologies include constructions where voltage ratings and interrupt